

Self Assessment:

1. Write a brief definition for "Artificial Intelligence"; from the Thinking Humanly view and Acting Humanly view

2. True or False:

- A. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- B. An agent's choice of action at any given instant can depend on the entire percept sequence observed to date, or on anything it hasn't perceived.
- C. It is possible for a known environment to be partially observable.
- D. The agent function maps any given percept sequence to an action; but the agent program is an implementation, running within some physical system.
- E. The notation "Task Environment" describe: performance, environment, actuators, and sensors of the agent.

Self Assessment:

1. Are the statements, “it will not rain or snow” and “it will not rain and it will not snow” logically equivalent

2. Show that

$$\frac{\begin{array}{c} P \rightarrow Q \\ \neg P \rightarrow Q \end{array}}{\therefore Q}$$

is a valid deduction rule?

3. Determine if the following is a valid deduction rule:

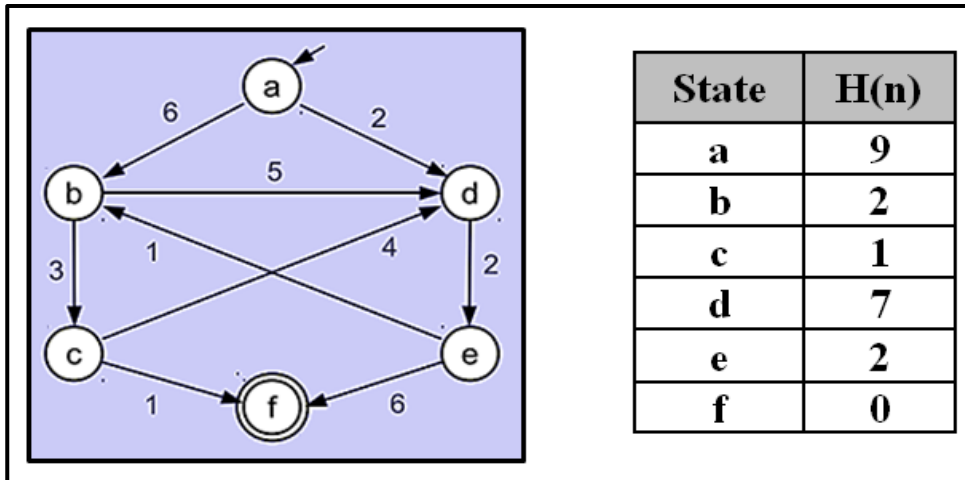
$$\frac{\begin{array}{c} P \rightarrow Q \\ \neg Q \end{array}}{\therefore P}$$

- 4. Are the following three statements logically equivalent?**

$(p \wedge \neg q) \vee (\neg p \wedge q),$
 $(\neg p \rightarrow q) \wedge (\neg (p \wedge q)),$
 $\neg ((p \wedge q) \vee ((\neg p) \wedge (\neg q))).$

1. Find the path from a to f using:

- A. Greedy Search; draw the entire traversal search tree.
- B. A* Tree Search.



2. Which solution would A* Tree Search (A* Search) find to move from node S to node G if run on the graph below?

